MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, HIGH SPEED TRACK AND HOLD AMPLIFIER, HYBRID, SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for high speed track and hold amplifiers. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	High speed track and hold amplifier

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
Y	(24-lead, .800" x 1.330") dual-in-line package, see figure 1
Z	(24-lead, .800" x 1.375") dual-in-line package, see figure 2

1.3 Absolute maximum ratings.

Positive supply voltage (V_{CC}) to common	+18 V dc
Negative supply voltage (V_{EE}) to common	-18 V dc
+5 volt supply voltage (V_1) to common	-0.5 V dc to +0.7 V dc
Analog input voltage range (pin 13)	$\pm V_{CC}$
Digital inputs (pins 11, 12)	-0.5 V dc to +5.5 V dc
Output current 1/	±50 mA
Operating case temperature range	-55°C to +125°C
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 60 seconds)	+300°C
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Positive supply voltage range (V_{CC})	+14.25 V dc to +15.75 V dc
Negative supply voltage range (V_{EE})	-14.25 V dc to -15.75 V dc
+5 volt supply (V_1)	+4.75 V dc to +5.25 V dc
Input voltage range	-10.0 V dc to +10.0 V dc
Output current	±20 mA
Maximum capacitive load	30 pF
Case temperature	-55°C to +125°C

1/ The duration of the output short circuit for any combination of supply voltage, output short current, and temperature shall not cause the junction temperature to exceed its maximum rating.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
24-lead DIP	Y	1.025 W at +125°C	50°C/W	70°C/W
24-lead DIP	Z	1.025 W at +125°C	50°C/W	70°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 3.

3.2.2 Functional schematic circuit. The functional schematic circuit shall be as specified on figure 4.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.2 herein.

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TABLE I. Electrical performance characteristics for device type 01.

Characteristic	Symbol	Conditions (see 3.4 and figure 7 unless otherwise specified)	Limits		Unit
			Min	Max	
V_{CC} supply current	I_{CC}	V_{IN} = 0 V V(pin 11) = 0 V V(pin 12) = 0 V	0	+30	mA
V_{EE} supply current	I_{EE}	V_{IN} = 0 V V(pin 11) = 0 V V(pin 12) = 0 V	-30	0	mA
V_{LOG} supply current	I_L	V_{IN} = 0 V V(pin 11) = 0 V V(pin 12) = 0 V	0	+25.0	mA
Total power consumption	P_T		0	975.0	mW
Logic input current (high)	I_{IH}	V_{IH} = 2.4 V	-0.1	60.0	µA
Logic input current (low)	I_{IL}	V_{IL} = 0.4 V	-2.4	0.1	mA
High level input voltage 1/	V_{IH}	V_{IN} = 0 V V_{IH} = 2.0 V	2.0	5.5	V
Low level input voltage 1/	V_{IL}	V_{IN} = 0 V V_{IL} = 0.8 V	0	0.8	V
Input offset voltage	V_{IO}	Initial @ T_C = +25°C -55°C ≤ T_C ≤ +125°C End point @ T_C = +25°C V(pin 13) = 0 V	-5.0 -20.0 -12.5	5.0 20.0 12.5	mV
Input resistance	R_i	V_{IN} = +10 V V(pin 11) = 0 V V(pin 12) = 0 V	0.75	2.00	k Ω
Gain error	A_E	Initial @ T_C = +25°C -55°C ≤ T_C ≤ +125°C End point @ T_C = +25°C	-0.1 -0.15 -0.2	+0.1 +0.15 +0.2	%
Gain linearity error	A_L	Best straight line (5 points)	-0.01	+0.01	% FS

See footnote at end of table.

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TABLE I. Electrical performance characteristics for device type 01 - Continued.

Characteristic	Symbol	Conditions (see 3.4 and figure 7 unless otherwise specified)	Limits		Unit
			Min	Max	
Power supply rejection ratio V_{CC}	$+P_{SRR}$	+14.25 V dc ≤ V_{CC} ≤ +15.75 V dc V_{IN} = 0 V V(pin 11) = 0 V V(pin 12) = 0 V	-1.0	+1.0	mV/V
Power supply rejection ratio V_{EE}	$-P_{SRR}$	-15.75 V dc ≤ V_{EE} ≤ -14.25 V dc V_{IN} = 0 V V(pin 11) = 0 V V(pin 12) = 0 V	-1.0	+1.0	mV/V
"Hold step" (pedestal voltage)	V_{HS}	T_C = +25°C	-20.0	+20.0	mV
Pedestal voltage temperature sensitivity	$\frac{\Delta V_{HS}}{\Delta T_C}$		-80.0	+80.0	µV/°C
Hold mode droop	V_{HD}	Initial @ T_C = +25°C End point @ T_C = +25°C	-5.0 -10.0	+5.0 +10.0	µV/us µV/us
		T_C = +125°C	-1.8	+1.8	mV/us
Track-to-hold transient	V_{TTH}	See figure 12	0	380	mVp-p
Acquisition time	t_{aq}	10 V step to ±1 mV See figure 8	0	200	ns
Transient response (settling time, track to hold)	$T_{R(ts)}$	Settling to ±1 mV	0	100	ns
		Settling to ±10 mV See figure 12	0	85	ns
Feedthrough rejection ratio	F_{RR}	V_{IN} = 20 Vp-p @ 2.5 MHz See figure 10	70		dB
Output resistance	R_O	See figure 7	0	0.25	ohm
Output slew rate	$\frac{\Delta V_O}{\Delta t}$	V_{IN} = -5 V to +5 V step V(pin 11) = 0 V V(pin 12) = 0 V See figure 12	120		V/µs

See footnote at end of table.

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TABLE I. Electrical performance characteristics for device type 01 - Continued.

Characteristic	Symbol	Conditions (see 3.4 and figure 7 unless otherwise specified)	Limits		Unit
			Min	Max	
Small signal bandwidth (-3 dB)	B_W	V_{IN} = 1 Vp-p ac V(pin 11) = 0 V V(pin 12) = 0 V See figure 13	8		MHz
Acquisition time	t_{aq}	10 V step to ±10 mV	0	170	ns
		1 V step to ±10 mV	0	100	ns
Aperture time	t_{ap}	Information only	0	16	ns
Aperture jitter	J_{ap}	Information only	-50	+50	ps

1/ If the manufacturer's automatic test equipment cannot adjust the logic input voltage thresholds (during dynamic tests) to the requirements of 3.4, herein, then the input voltage threshold test shall be run as a separate parameter.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified. The following conditions shall also apply unless otherwise specified:

V_{CC} = +15.0 V
V_{EE} = -15.0 V
V_{LOG} = +5.0 V
V_{IL} = +0.8 V
V_{IH} = +2.0 V
-55°C $\leq T_C \leq$ +125°C

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests subgroups 1 and 4 specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit technology group I (see MIL-M-38510, appendix E).

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modifications to correction and proof test power supplies

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D) using the circuit shown on figure 6. Test duration shall be as defined in MIL-STD-883.

(2) Class B devices: Test condition D, using the circuit shown on figure 6, or test condition C, using the circuit shown on figure 5.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Internal visual inspection shall be in accordance with method 2017 of MIL-STD-883.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883, and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883, and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 9, 10, and 11 shall be omitted.

c. A special subgroup shall be added to group A inspection for class S only and it shall consist of the conditions and limits as specified in table III. Subgroup 12, LTPD shall be 15.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using a circuit shown on figure 6.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1,2,3,4,5,6	1,4,5,6
Group A test requirements (method 5005)	1,2,3,4,5, 6,7,8A,8B, 12	1,2,3,4,5,6, 7,8A,8B
Group C end-point and group B, class S electrical parameters (method 5005)	1, 2, 3, 4, and table IV delta limits	1, 4, and table IV delta limits
Additional electrical subgroups for group C periodic inspection	N/A	12
Group D end-point electrical parameters (method 5005)	1, 2, 3	1

1/ *PDA applies to subgroup 1 (see 4.2c).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition C, using the circuit shown on figure 5; or test condition D, using the circuit shown on figure 6.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. A special subgroup shall be added to the group C inspection for class B only requirements and shall consist of the tests, conditions, and limits specified for subgroup 12 of group A. The LTPD for the special subgroup shall be 15.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. This group shall be measured only for initial qualification and thereafter for group D tests at +25°C.

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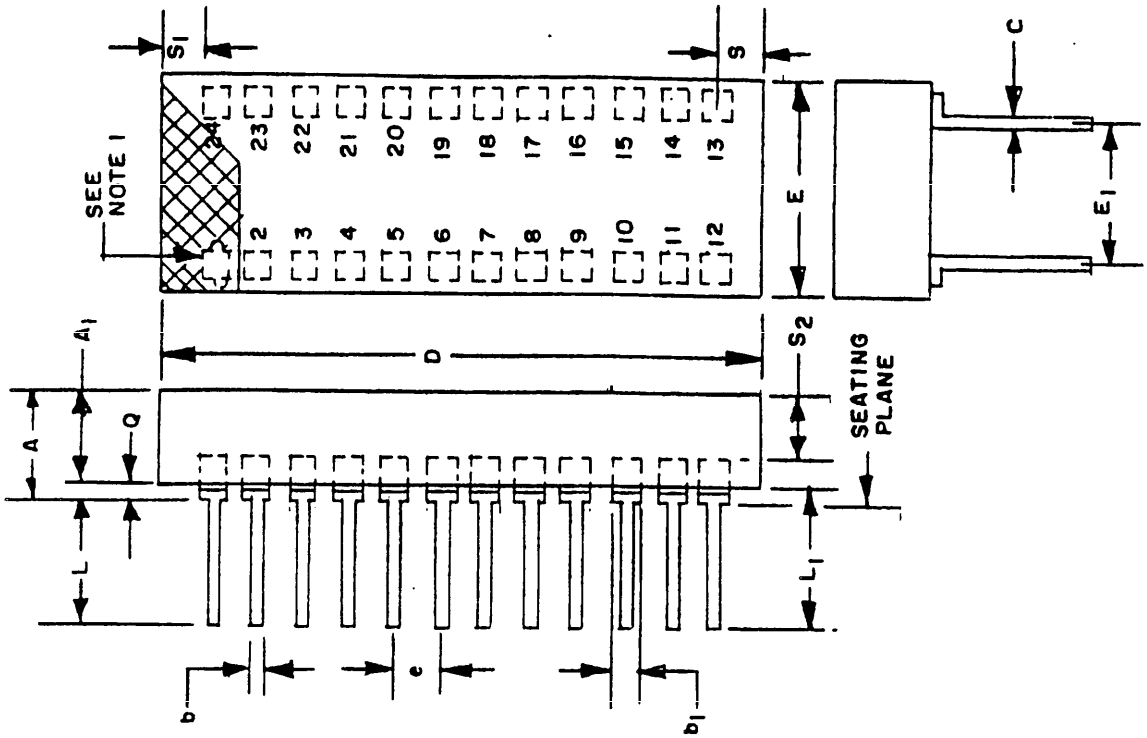


FIGURE 1. Case outline Y.

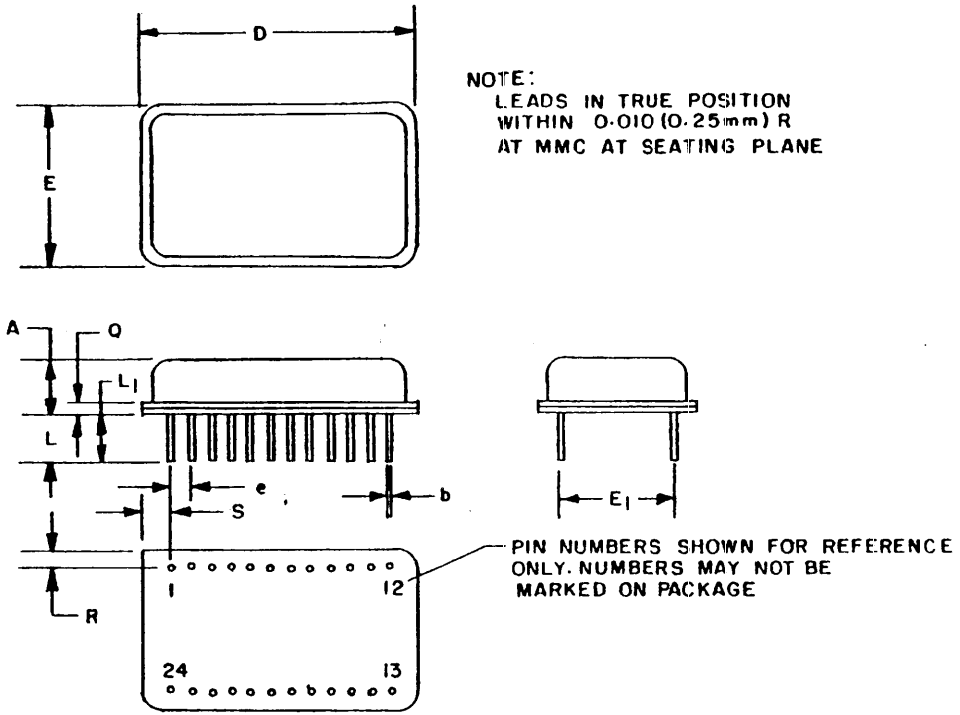
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A		.236		5.99	
A_1	.120	.170	3.05	4.32	
b	.016	.022	0.40	0.55	8
b_1	.035	.045	0.89	1.14	2,8
C	.009	.015	0.23	0.38	8
D	1.275	1.385	32.39	35.18	4
E	.770	.830	19.56	21.08	4
E_1	.590	.610	14.99	15.49	7
e	.100 BSC		2.54 BSC		9
L	.170	.215	4.32	5.46	
L_1	.205	.250	5.21	6.35	
Q	.015	.035	0.38	0.89	3
S	.087	.125	2.21	3.18	6
S_1		.114		2.90	6
S2	.010				

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NOTES:
1. Index area; a notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension b_1 may be .023 (0.58 mm) for leads number 1, 12, 13, and 24 only.
3. Dimension Q shall be measured from the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus and glass overrun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (0.25 mm) of its exact longitudinal position relative to pins 1 and 24.
6. Applies to all four corners (leads number 1, 12, 13, and 24).
7. L_1 shall be measured at the centerline of the leads.
8. All leads - increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when finish A or B is applied.
9. Twenty-two spaces.
10. If this configuration is used, no organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
11. Dimensions are in inches.
12. Metric equivalents are given for general information only.

FIGURE 1. Case outline Y - Continued.

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Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.170	.250	4.318	6.35	
b	.016	.021	.406	.533	7
D	1.365	1.385	34.67	35.18	3
E	.790	.810	20.07	20.57	3
E1	.590	.610	14.99	15.49	6
e	.100	BSC	2.54	BSC	4,8
L		.285	3.81	7.62	
L1	.150	.300			
Q		.015			
R	.080	.110	2.032	2.794	5
S	.125	.150	3.175	3.81	5

NOTES:
1. The manufacturer's identification shall not be used as a pin one identification mark.
2. Dimension Q shall be measured from the seating plane to the base plane.
3. This demension allows for off-center lid overrun.
4. The basic pin spacing is .100 (2.54 mm) between center lines. Each pin center line shall be located within .010 (0.25 mm) of its exact to pins 1 and 24.
5. Applies to all four corners (leads 1, 12, 13, 24).
6. E_1 shall be measured at the center line of the leads.
7. All leads - increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A or B is applied.
8. Twenty-two spaces.
9. Dimensions are in inches.
10. Metric equivalents are given for general information only.

FIGURE 2. Case outline Z.

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Pin	Function	Pin	Function
1	Analog output	13	Analog input
2	N.C.	14	N.C.
3	N.C.	15	Input ground
4	N.C.	16	N.C.
5	N.C.	17	N.C.
6	N.C.	18	N.C.
7	N.C.	19	N.C.
8	N.C.	20	N.C.
9	+5V (V_1)	21	Ground
10	Logic ground	22	-15V (V_{EE})
11	Hold	23	Power ground
12	Hold	24	+15V (V_{CC})

FIGURE 3. Terminal connections.

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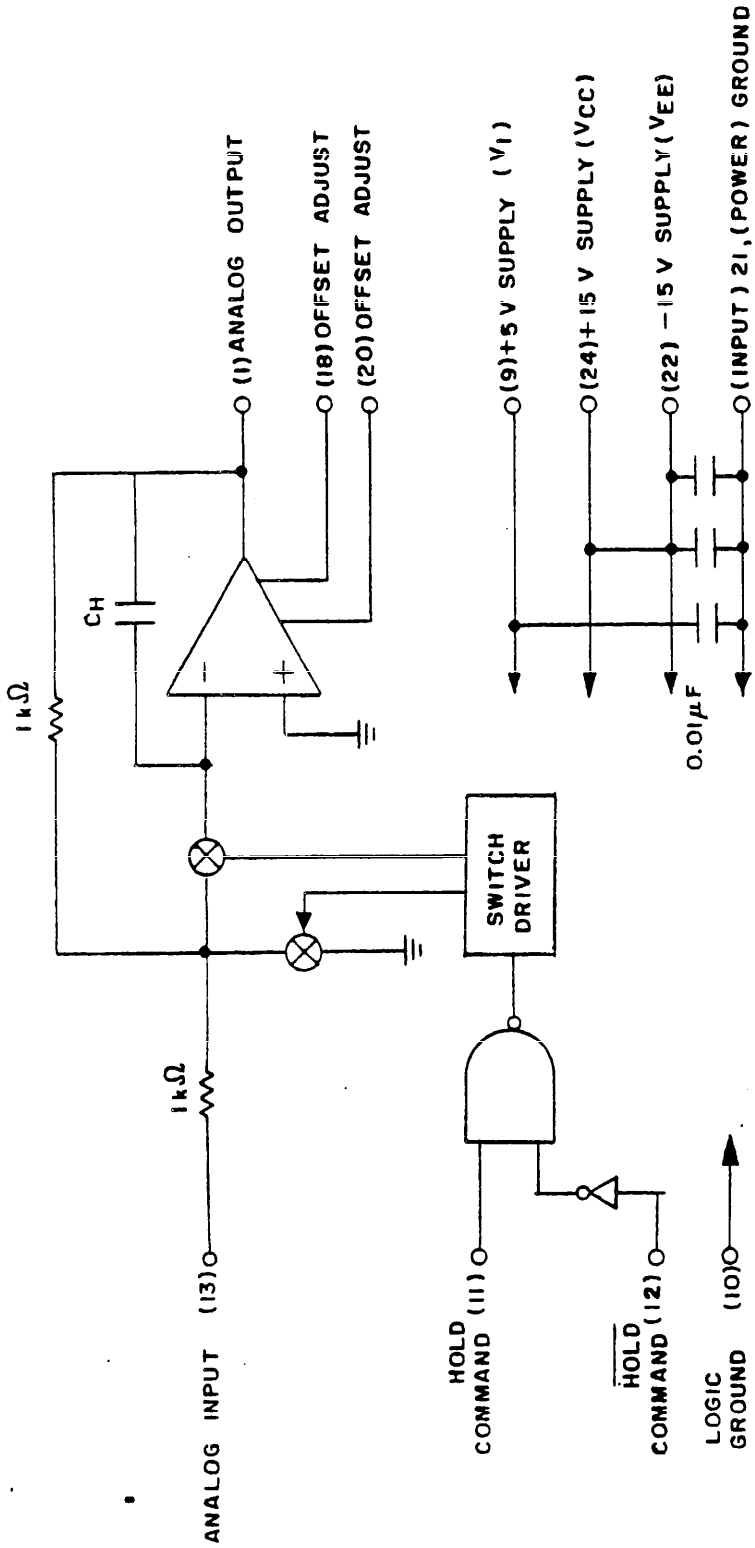
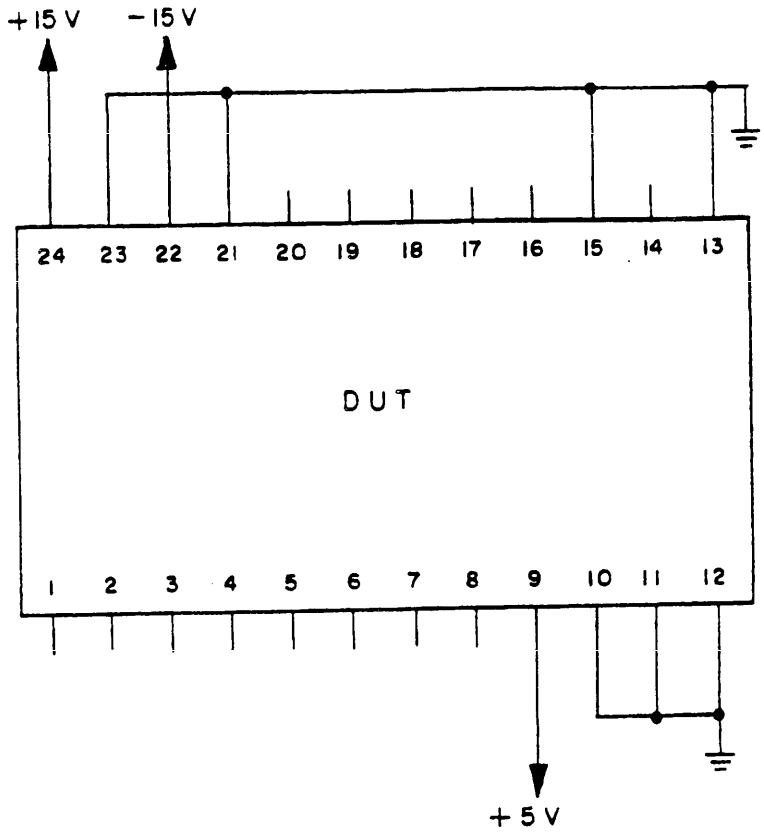


FIGURE 4. Functional schematic circuit.

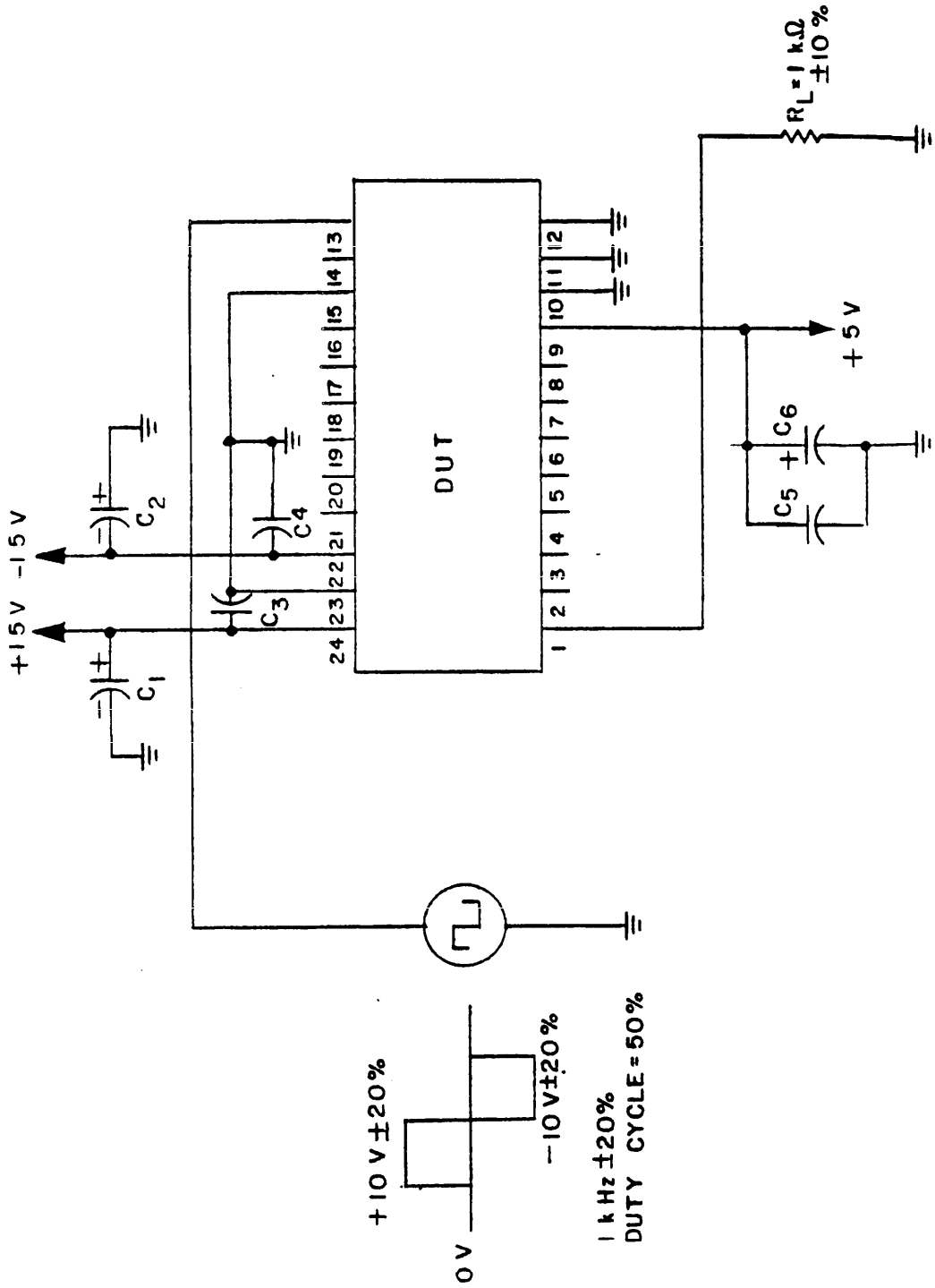
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NOTES:
1. Optional filter ($\geq$ 1.0 µF) and bypass ($\leq$ 0.1 µF) capacitors may be added to each supply line.
2. T_C = +125°C.

FIGURE 5. Static burn-in and life test circuit.

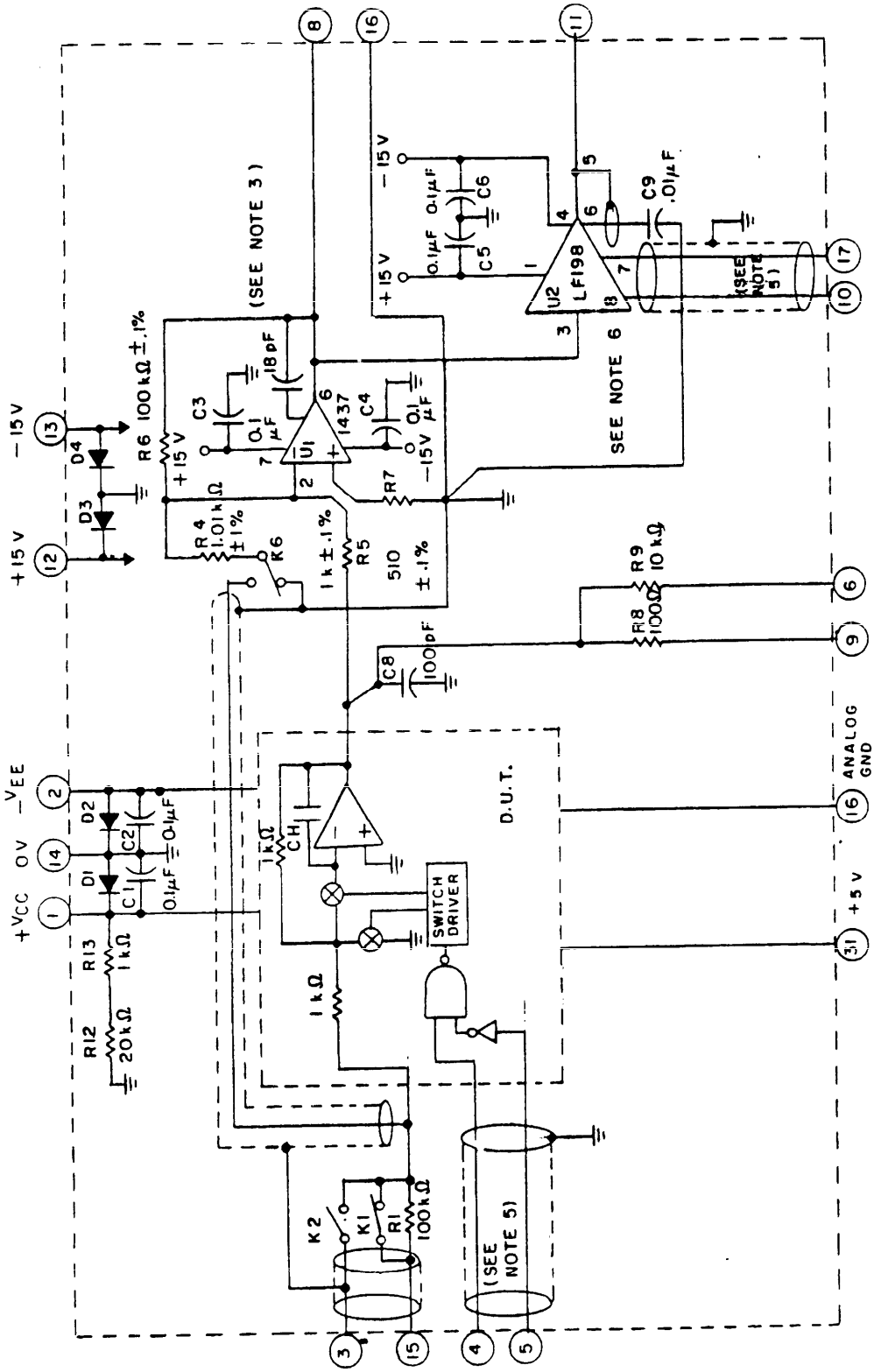
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NOTE: Optional filter ($\geq$ 1.0 µF) and bypass ($\leq$ 0.1 µF) capacitors may be added to each supply line.

FIGURE 6. Dynamic burn-in and life test circuit.

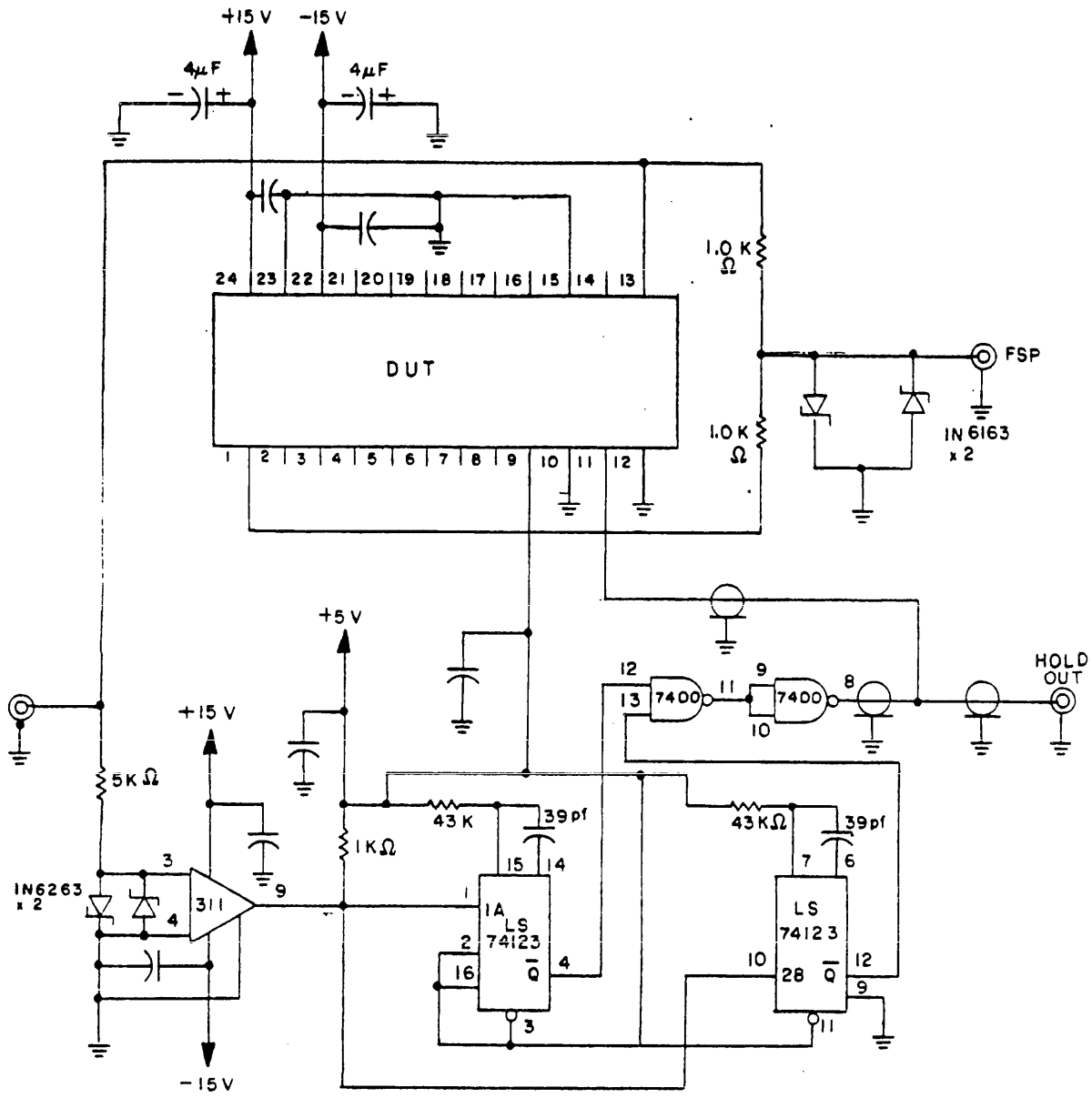
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NOTES:
1. Last component designations are R13, C9, U2, D4 and K8.
2. All resistors are ± 1 percent film unless otherwise stated.
3. U1 offset error shall be measured with Pin 9 grounded and the D.U.T. removed. This error shall be removed in the software calculations.
4. Relay control inputs are not shown.
5. A speed-up circuit is required in series with the logic input for rise times greater than 0.5 μs.
6. The adapter S/H U2 is required for T_C = 125°C testing of some parameters.

FIGURE 7. Test circuit for static and dynamic tests.

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NOTES:
1. The acquisition time is measured from the time the Hold out signal goes low to when the output signal at the FSP output is within ± 0.01 percent, (± 1.0 mv).
2. All capacitors are 0.1 μF unless otherwise stated.

FIGURE 8. Acquisition time test circuit.

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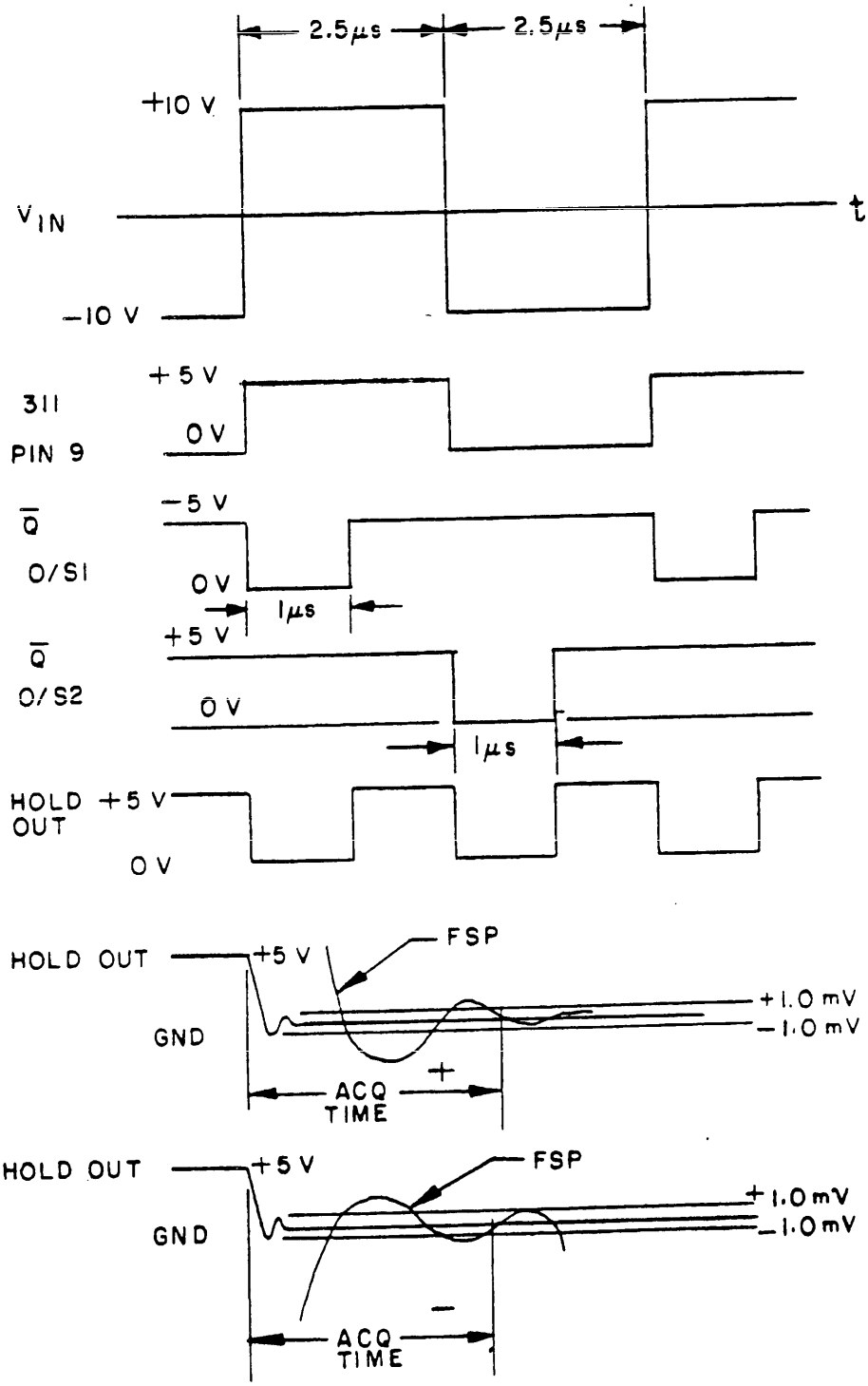
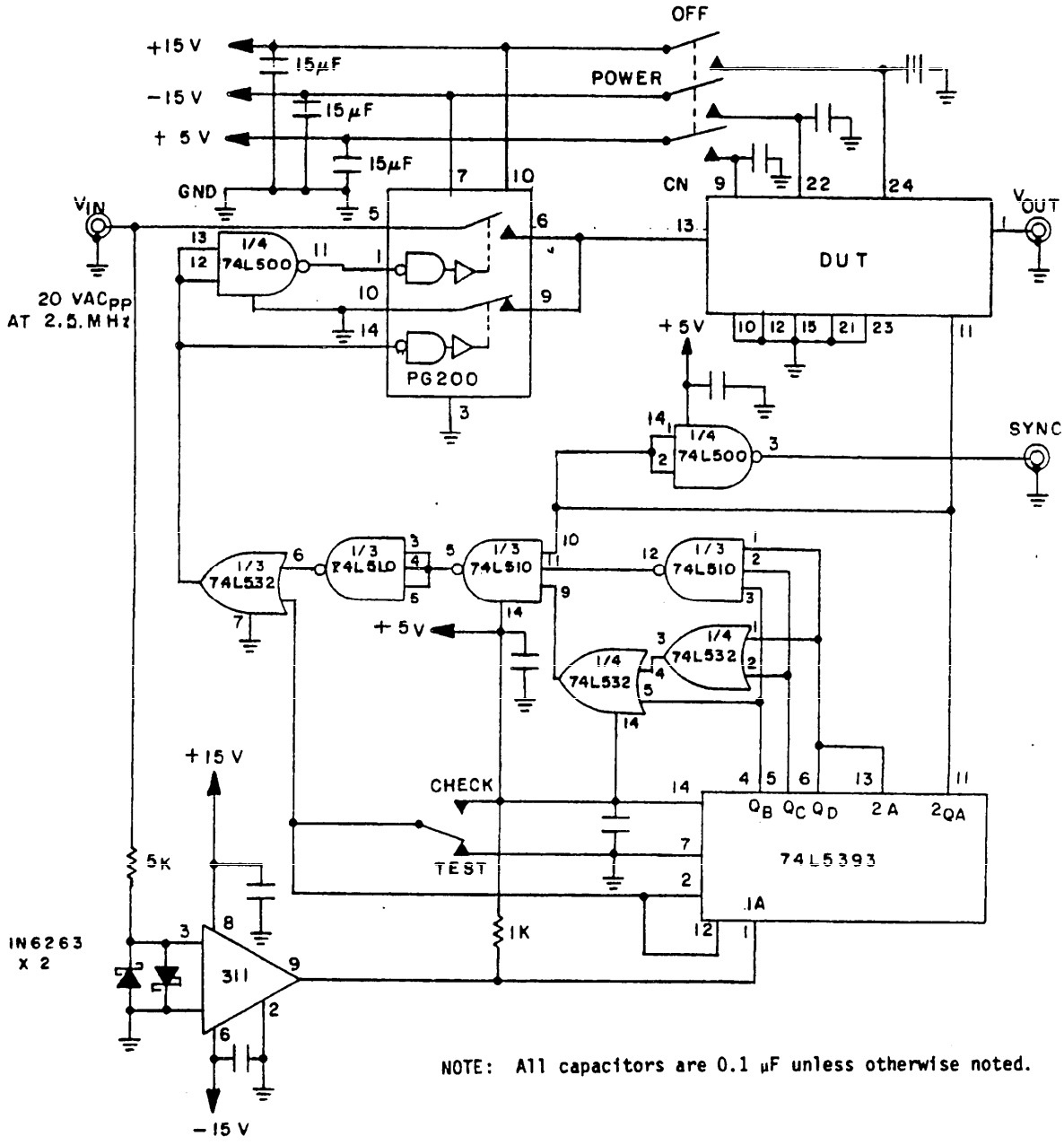


FIGURE 9. Acquisition time test circuit Continued .

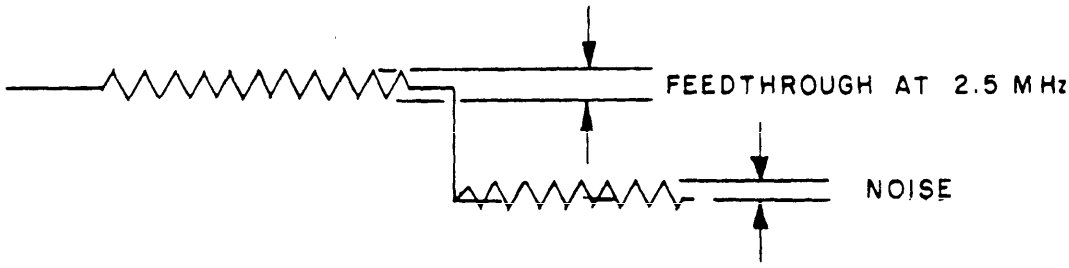
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NOTE: All capacitors are 0.1 μF unless otherwise noted.

FIGURE 10. AC feedthrough rejection ratio test circuit.

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$$\text{FEEDTHROUGH}_{ac} = 20 \text{ LOG} \frac{20000}{V_o \text{ (mV)}} \text{ (dB)}$$

FIGURE 11. AC feedthrough rejection ratio test circuit waveforms.

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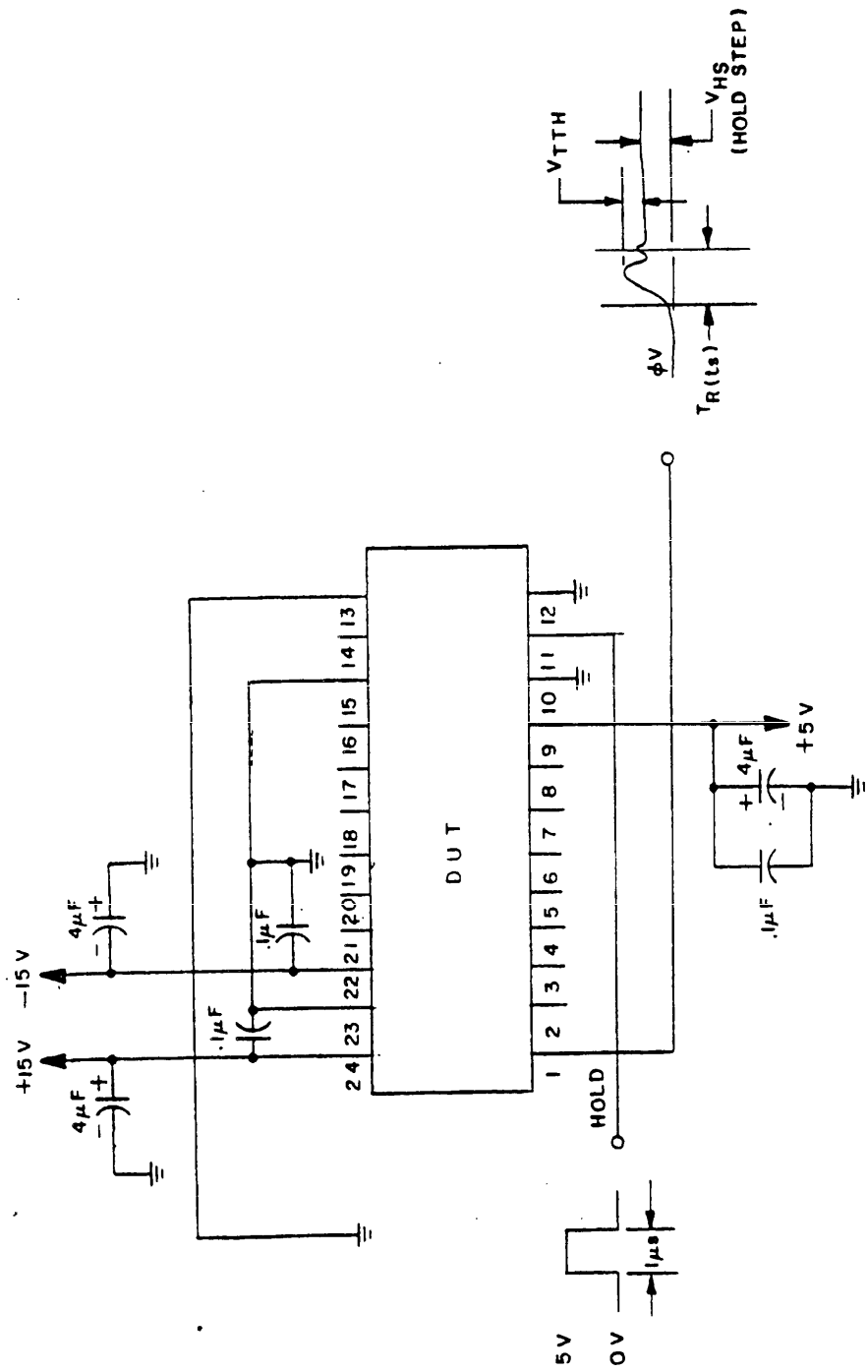


FIGURE 12. Track to hold settling time test circuit and waveforms.

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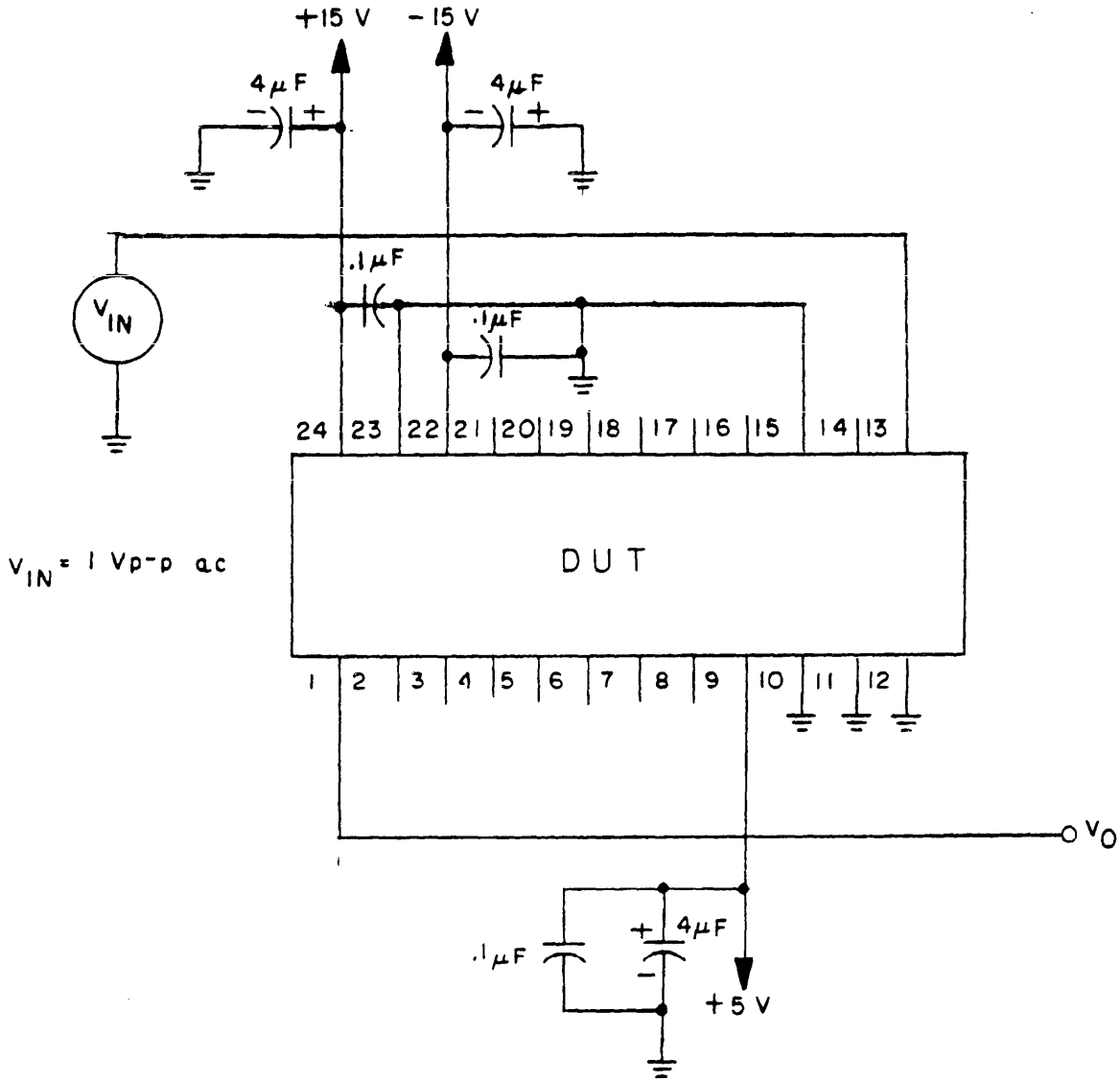


FIGURE 13. Small signal bandwidth test circuit.

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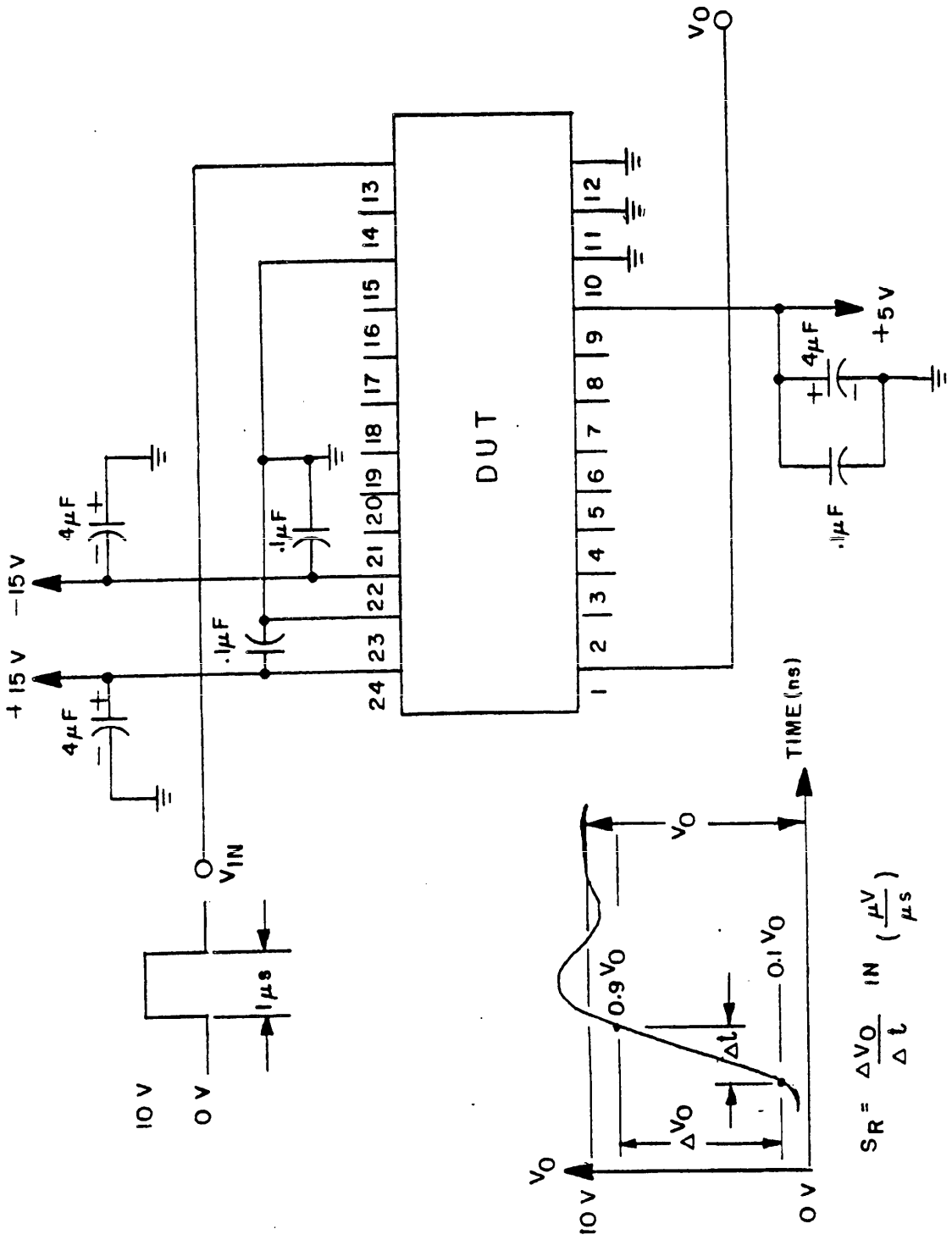
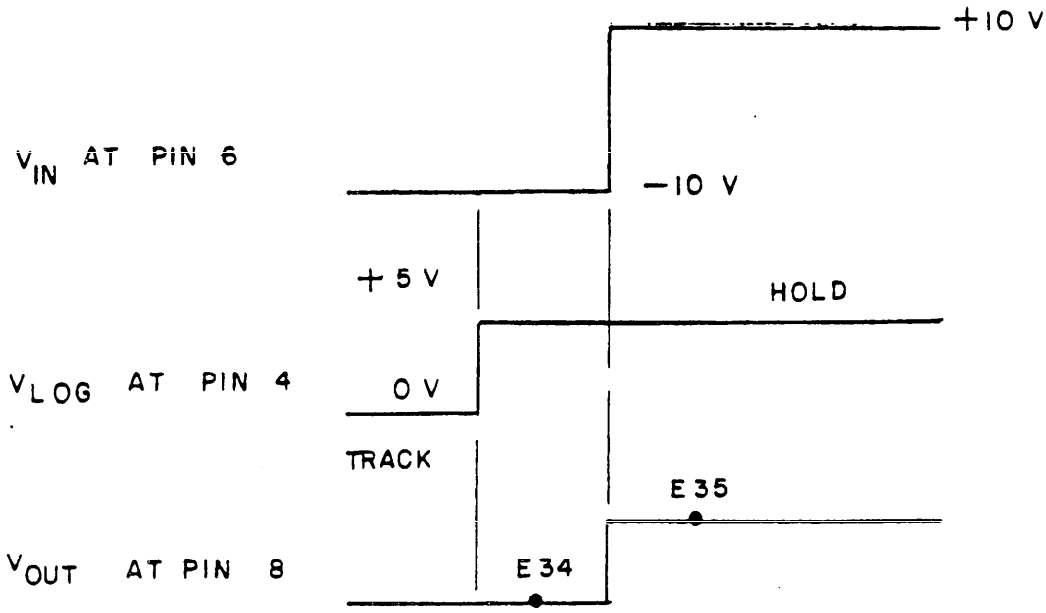


FIGURE 14. Slew rate test circuit and waveforms.

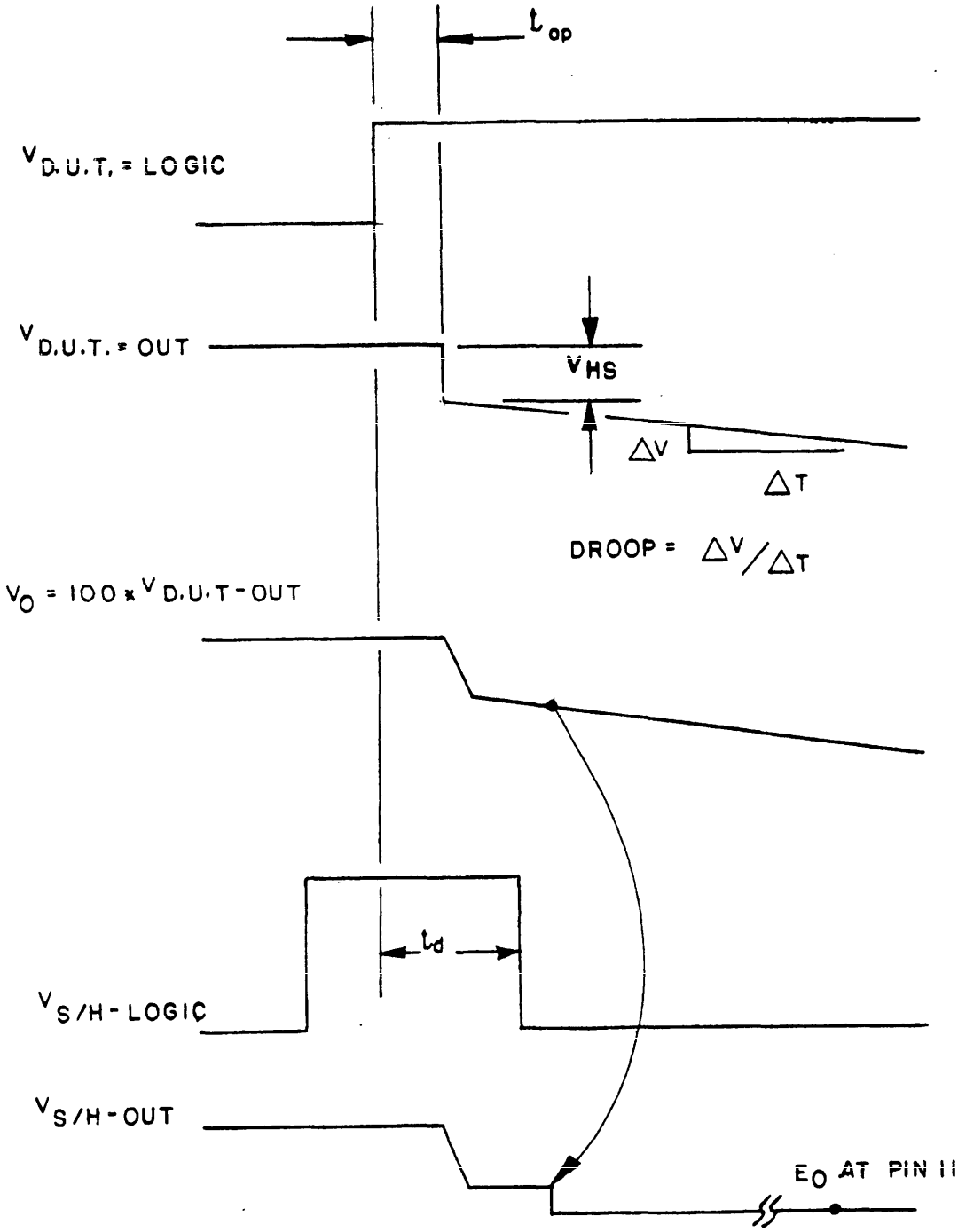
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$R_O = 5\ (E35 - E34)\ @\ T_C = +25°C$
$R_O = 5\ (E38 - E37)\ @\ T_C = +125°C$
$R_O = 5\ (E40 - E39)\ @\ T_C = -55°C$

FIGURE 15. Output impedance test timing.

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NOTES:
1. At T_C = + 125°C hold mode leakage current causes excessive droop rate of the hold capacitor voltage.
2. In order to minimize the droop error, the test fixture S/H at T_C =+125°C is used to acquire and hold the voltage until the measurement system can do its function.
3. See figure 7 test circuit.

FIGURE 16. Hold mode droop measurements at T_C = +125°C.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	MIL STD-883 method	Test no.	Notes	Adapter pin numbers									Energized relays	Measured pin			Equations 1/	Limits		Units
					1	2	3	4	5	6	10	17	31		No.	Value	Unit		Min	Max	
Calibration			1	2/	Apply 0 volts to pin 9 with OUT removed.									NONE	8	E1	V	V_{CAL} = c1/200			mV
1/ T_C = +25°C	I_{CC}	5005	2		+15 V	-15 V	0 V	0 V	0 V	OPEN	OPEN	OPEN	5 V	"	1	I1	mA	I_{CC} = I1	0	+30	mA
	I_{EE}	"	3		"	"	"	"	"	"	"	"	"	"	2	I2	"	I_{EE} = I2	-30	0	"
	I_I	"	4		"	"	"	"	"	"	"	"	"	"	31	I3	"	I_I = I3	0	+25	"
	P_T	"	5	P_T is calculated using data from tests 1, 2, and 3.														P_T = 15 (I1 + I121) + 513	0	975	mW
	I_{IH}	"	6		+15 V	-15 V	0 V	0 V	0 V	OPEN	OPEN	OPEN	5 V	NONE	4	I4	µA	I_{IH} = I4	-0.1	60	µA
	I_{IL}	"	7		"	"	"	"	"	"	"	"	"	NONE	4	I5	mA	I_{IL} = I5	-2.4	+0.1	mA
	V_{IO}	"	8		"	"	"	"	"	"	"	"	"	K6	8	E2	V	V_{IO} = (2E1 - E2)/100	-5	+5	mV
	V_{IL}	"	9	3/	"	"	"	"	"	"	"	"	"	"	4	E3	"	V_{IL} = E3	0	0.8	V
	V_{IH}	"	10	4/	"	"	"	"	"	"	"	"	"	"	4	E4	"	V_{IH} = E4	2.0	5.5	V
	R_I	"	11		"	"	10 V	"	"	"	"	"	"	"	3	I6	mA	R_I = 10/I6	0.75	2.0	kΩ
	A_E	"	12		"	"	10 V	"	"	"	"	"	"	"	8	E5	V	A_E = (E5 - E2)/10	-0.1	+0.1	%
	A_L	"	13	5/	"	"	5/	"	"	"	"	"	"	"	"	$E_{(1)}$	"	A_L = 10 ($mE_{r(1)}$ + b - $E_{(1)}$)	-0.01	0.01	%
	$+P_{SRR}$	"	14		"	-15.75 V	0 V	"	"	"	"	"	"	"	"	E6	"	$+P_{SRR}$ = (E6 - E7)/1.5	-1.0	+1.0	mV/V
			15		"	-14.25 V	"	"	"	"	"	"	"	"	"	E7	"				
	$-P_{SRR}$	"	16		+15.75 V	-15 V	"	"	"	"	"	"	"	"	"	E8	"	$-P_{SRR}$ = (E8 - E9)/1.5	-1.0	+1.0	mV/V
			17		+14.25 V	"	"	"	"	"	"	"	"	"	"	E9	"				
2/ T_C = +125°C	I_{CC}	"	18		+15 V	"	"	"	"	"	"	"	"	NONE	1	I7	mA	I_{CC} = I7	0	+30	mA
	I_{EE}	"	19		"	"	"	"	"	"	"	"	"	"	2	I8	"	I_{EE} = I8	-30	0	"
	I_I	"	20		"	"	"	"	"	"	"	"	"	"	31	I9	"	I_I = I9	0	+25	"
	P_T	"	21	P_T is calculated using data from tests 18, 19, and 20.														P_T = 15 (I7 + I181) + 519	0	975	mW
	I_{IH}	"	22		+15 V	-15 V	0 V	0 V	0 V	OPEN	OPEN	OPEN	5 V	NONE	4	I10	µA	I_{IH} = I10	-0.1	60	µA
	I_{IL}	"	23		"	"	"	"	"	"	"	"	"	NONE	4	I11	mA	I_{IL} = I11	-2.4	+0.1	mA
	V_{IO}	"	24		"	"	"	"	"	"	"	"	"	K6	8	E10	V	V_{IO} = (2E1 - E10)/100	-20	+20	mV
	V_{IL}	"	25	3/	"	"	"	"	"	"	"	"	"	"	4	E11	"	V_{IL} = E11	0	0.8	V
	V_{IH}	"	26	4/	"	"	"	"	"	"	"	"	"	"	4	E12	"	V_{IH} = E12	0	2.0	V
	R_I	"	27		"	"	10 V	"	"	"	"	"	"	"	3	I12	mA	R_I = 10/I12	0.75	2.0	kΩ

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	10	17	31	Energized relays	Measured pin No.	Value	Unit	Equations 1/	Limits Min	Max	Units
2 T_C = +125°C	A_E	5005	28		+15 V	-15 V	10 V	0 V	0 V	OPEN	OPEN	OPEN	5 V	K6	8	E13	V	A_E = (E13 - E2)/10	-0.15	+0.15	%
	A_L	"	29	5/	"	-15 V	5/	"	"	"	"	"	"	"	"	E(1)	"	A_L = 10 ($mE_{r(1)}$ + b - E(1))	-0.01	0.01	%
	$+P_{SRR}$	"	30		"	-15.75 V	0 V	"	"	"	"	"	"	"	"	E14	"	$+P_{SRR}$ = (E14 - E15)/1.5	-1.0	+1.0	$\frac{mV}{V}$
			31		"	-14.25 V	"	"	"	"	"	"	"	"	"	E15	"				
	$-P_{SRR}$	"	32		+15.75 V	-15 V	"	"	"	"	"	"	"	"	"	E16	"	$-P_{SRR}$ = (E16 - E17)/1.5	-1.0	+1.0	$\frac{mV}{V}$
			33		+14.25 V	-15 V	"	"	"	"	"	"	"	"	"	E17	"				
3 T_C = -55°C	(All test characteristics, conditions, equations, and test limits are identical with those specified in table III, subgroup 2, T_C = +125°C, test numbers 18 through 33.)																				
4 T_C = +25°C	V_{HS}	5005	50		+15 V	-15 V	See figure 12			OPEN	OPEN	OPEN	5 V	K6	8	E26	V	V_{HS} = (E26 - E2)/100	-20	+20	mV
	V_{HD}	"	51		"	"				"	"	"	"	"	"	E27	"	V_{HD} = (E28 - E27)/Δt	-5	+5	$\frac{\mu V}{\mu s}$
			52		"	"				"	"	"	"	"	"	E28	"				
	V_{TTH}	"	53		"	"	See figure 12			"	"	"	"	"	"	E29	"	V_{TTH} = E29/100	0	380	mV p-p
	t_{aq}	"	54	6/	"	"	See figure 9			"	"	"	"	"	"	t1	ns	t_{aq} = t1	0	200	ns
5 T_C = +125°C	V_{HD}	"	55	7/	"	"	See figure 16			"	0 V	0 V	"	"	11	E30	V	V_{HD} = (E31 - E30)/Δt	-1.8	+1.8	$\frac{mV}{\mu s}$
			56	7/	"	"				"	0 V	0 V	"	"	11	E31	"				
	V_{TTH}	"	57		"	"	See figure 12			"	OPEN	OPEN	"	"	8	E32	"	V_{TTH} = E32/100	0	380	mVp-p
6 T_C = -55°C	V_{TTH}	"	58		"	"	See figure 12			"	"	"	"	"	"	E33	"	V_{TTH} = E33/100	0	380	mVp-p
7 T_C = +25°C	R_D	"	59		"	"	See figure 15			+10 V	0 V	0 V	"	"	"	E34	V	R_D = 5 (E35 - E34)	0	0.25	Ω
			60		"	"				-10 V	"	"	"	"	"	E35	"				
	$F_{RR(AC)}$	"	61		"	"	See figure 11			OPEN	"	"	"	"	"	E36	"	$F_{RR(AC)}$ = 20 LOG (20000/E36)	70		dB
	$T_{R(ts)}$	"	62	8/	"	"	See figure 12			"	OPEN	OPEN	"	"	"	t2	ns	$T_{R(ts)}$ = t2	0	100	ns
	t_{aq}	"	63	9/	"	"	See figure 9			"	"	"	"	"	"	t3	"	t_{aq} = t3 V_{IN} = -10 V to 0 V	"	200	"
	t_{aq}	"	64	"	"	"	"			"	"	"	"	"	"	t4	"	t_{aq} = t4 V_{IN} = 0 V to -10 V	"	"	"
	t_{aq}	"	65	"	"	"	"			"	"	"	"	"	"	t5	"	t_{aq} = t5 V_{IN} = +10 V to 0 V	"	"	"
	t_{aq}	"	66	"	"	"	"			"	"	"	"	"	"	t6	"	t_{aq} = t6 V_{IN} = 0 V to +10 V	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	10	17	31	Energized relays	Measured pin No.	Value	Unit	Equations 1/	Limits Min	Max	Units
8A T_C = +125°C	R_O	5005	67	7/	+15 V	-15 V	See figure 15			+10 V	0 V	0 V	5 V	K6	11	E37	V	R_O = 5(E38 - E37)	0	0.25	Ω
			68	"	"	"				-10 V	"	"	"	"	"	E38	"				
	$F_{RR(AC)}$	"	69	"	"	"	See figure 11			OPEN	"	"	"	"	"	E39	"	$F_{RR(AC)}$ = 20 LOG (20000/E39)	70		dB
	$T_{R(ts)}$	"	70	8/	"	"	See figure 12			"	OPEN	OPEN	"	"	8	t7	ns	$T_{R(ts)}$ = t7	0	100	ns
	t_{aq}	"	71	9/	"	"	See figure 9			"	"	"	"	"	"	t8	"	t_{aq} = t8 V_{IN} = -10 V to 0 V	"	200	"
	t_{aq}	"	72	"	"	"	"			"	"	"	"	"	"	t9	"	t_{aq} = t9 V_{IN} = 0 V to -10 V	"	"	"
	t_{aq}	"	73	"	"	"	"			"	"	"	"	"	"	t10	"	t_{aq} = t10 V_{IN} = +10 V to 0 V	"	"	"
	t_{aq}	"	74	"	"	"	"			"	"	"	"	"	"	t11	"	t_{aq} = t11 V_{IN} = 0 V to +10 V	"	"	"
8B T_C = -55°C	(All test characteristics, conditions, equations, and test limits are identical with those specified in table III, subgroup 8A, T_C = +125°C, test numbers 67 through 74.)																				
12 T_C = +25°C	B_W	5005	83	10/	+15 V	-15 V	See figure 13			OPEN	OPEN	OPEN	5 V	K6	8	E43	V	B_W = E43	8		MHz
	S_R	5005	84		"	"	See figure 14			"	"	"	"	"	"	E44	"	S_R = (E45 - E44)/at	120		$\frac{\mu V}{\mu s}$
			85		"	"				"	"	"	"	"	"	E45	"				

1/ The equations take into account the test amplifier gain of 100 and other constants so that the calculated value is in the table I units.

2/ In order to remove test amplifier offset from the data values, measure the offset of U1 on pin 8 with pin 9 grounded and the DUT removed. Software substraction techniques shall be used to correct the data.

3/ V_{IL} is the maximum logic low level which guarantees the track and hold is in the track mode.

4/ V_{IH} is the minimum logic high level which guarantees the track and hold is in the hold mode.

5/ Nonlinearity error is defined in reference to a best straight line using the least squares method. An input voltage reference (Vr) shall be varied between -10 volts and +10 volts at 5 V increments. At each referenced Vr(i), the output voltage Vo(i) shall be measured and recorded. Computation of the best straight line using the least squares method is defined as:

$$m = \frac{nE(Vr(i)\ Vo(i)) - (EVo(i))(EVr(i))}{nE(Vr(i)^2) - (EVr(i))^2}$$

and $$b = \frac{EVo(i)}{n} - \frac{mEVr(i)}{n}$$

where n = number of data points

The best straight line is then defined as:

$$Vo(i) = mVr(i) + b$$

Nonlinearity error defined in reference to the best straight line using the least squares method is calculated by

$$\text{Nonlinearity} = \frac{Vo(i) - (mVr(i) + b)}{FSR}$$

where FSR is the full scale range; 20 V

6/ The acquisition time measured for this test is chosen from subgroup 7, test numbers 63 through 66.

7/ Hold mode droop at T_C = +125°C due to high leakage current will tend to mask the data for V_{HS}, R_O, and F_{RR}. To preclude this effect from happening, the test adapter sample/hold circuit, which is not at an elevated temperature should be used to acquire and hold data for the measurement system.

8/ Settling time is determined as shown on figure 12 as the time for the track/hold to settle to within 1 mV of the final value after the hold mode command is given.

9/ Not tested if tested in subgroup 4.

10/ Small signal bandwidth is determined by applying a 1 Vp-p ac signal at the track/hold input and increasing the frequency until the output is -3 dB (0.707 V). The bandwidth is that frequency at which the output voltage is down 3 dB from the input.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

4.5.2 Burn-in and life test cooldown procedures. When these tests are completed, and prior to removal of the bias voltages, the devices under test (DUT) shall be cooled to within 10°C of the power stable condition at 25°C ambient, and the electrical parameter end-point measurements shall then be performed.

TABLE IV. Group C end-point electrical parameters/delta limits at T_C = +25°C.

Table III test no.	Device type	Symbol	Delta 1/		End-point limits		Unit
			Min	Max	Min	Max	
8	01	V_{IO}	-7.5	+7.5	-12.5	+12.5	mV
12	01	A_E	-0.1	+0.1	-0.2	+0.2	%
51,52	01	V_{HD}	-5.0	+5.0	-10.0	+10.0	μV/μS

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	MN376H/Micro Networks
01	SHM-4860MM/GE/Datel
01	THA-05203-1/ILC-DDC
01	4860-83/Teledyne Philbrick
01	HTC-0300/Analog Devices

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/142	Microcircuits, Linear, High Speed Track and Hold Amplifier, Hybrid, Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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